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"I SUPPORT THEM, BUT…": PERCEPTIONS OF TRANSGENDER ATHLETES DEPEND ON ATHLETE GENDER AND SKILL LEVEL

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Abstract – We investigated perceptions of transgender athletes using an experimental design that manipulated descriptions of athletes' gender and ability level. We also examined whether characteristics of the perceiver would influence attitudes toward transgender athletes. Participants were provided with a narrative describing either a transgender male or female athlete who played on a college basketball team at either a low or high skill level. Participants provided their perceptions and responded to scales to assess their empathy, belief in a just world, and demographics. Participants expressed general support for transgender athletes and an awareness that these athletes face prejudice and discrimination. Matching our hypothesis, the transgender man with a high level of ability elicited the most support, while the transgender woman with a high level of ability elicited the lowest levels of support. Qualitative data revealed more mixed feelings than the quantitative data, with a focus on biological advantages that accompany male puberty. Participants who believed in a just world, were politically conservative, were lower in empathy, and older reported less support for transgender athletes than their counterparts. These findings suggest that perceptions of transgender athletes are based on the athletes' gender identity and performance, as well as characteristics of the perceiver.

The inclusion of transgender athletes in competitive sports is increasing in frequency and has resulted in social controversy (Reynolds & Hamidian, 2021; Safer, 2022). In 2020, the first legislation that banned transgender girls from school-sponsored female sports teams was passed in the United States (Martin & Rahilly, 2023). By 2023, 21 states established laws that prevent transgender students from participating in school sports that are incongruent with their sex at birth (Jakubowska, 2024). Attitudes surrounding the issue are complex and influential in driving policies; however, research on the topic is limited (Avalos et al., 2022; Kavoura & Kokkonen, 2021).

A common argument against transgender athletes competing on teams that match their gender identity is that men have a biological advantage over women due to their physiology and higher testosterone levels (Martin & Rahilly, 2023; Pigozzi et al., 2022). This gender disparity becomes more pronounced following puberty, as many of the changes resulting from male hormones persist over time (Hilton & Lundberg, 2021; Jakubowska, 2024). For example, following puberty, a typical male body will have greater cardiorespiratory capacity and muscle mass than a typical female body (Heather, 2022). Consequently, policies often factor in the developmental stage when the public gender transition takes place, the length of time an individual has undergone hormone replacement, or the length of time an individual has undergone suppression treatment (Jakubowska, 2024). As one example, the National Collegiate Athletic Association currently requires transgender athletes to provide documentation of testosterone levels prior to and during their sports season.

Because the male body is perceived to have an advantage over the female body, the extension of this argument is that banning transgender women, who have gone through male puberty, physically protects cisgender women and promotes fairness in competition (Martin & Rahilly, 2023). This argument relies heavily on the dichotomy reflected in sex-based chromosomes (Jakubowska, 2024), as well as performance records that consistently show superior performance by male athletes compared to female athletes in identical Olympic events (Hissa & Hissa, 2023). A counterargument against the emphasis on biology is an emphasis on sociocultural factors that contribute to identity and athleticism (Jakubowska, 2024). A related argument against

dichotomous sex categories is that levels and fluctuations of naturally occurring hormones can vary widely across developmental time and social situations (Parks, 2016).

Research on content analysis of legislative discourse uncovered persistent themes on both sides of the argument. Arguments against transgender inclusion aligned more closely with politically conservative ideology and reflected values of elitism, tradition, fairness, safety and equal opportunity (Flores et al., 2020; Martin & Rahilly, 2023; Jakubowska, 2024). Arguments supporting transgender inclusion aligned more closely with politically liberal ideology and focus more on human rights and gender fluidity (Moyer, 2022; Posbergh, 2023).

Although support for transgender athletes is increasing, the overriding public perception remains negative, particularly in certain societal groups (Avalos et al., 2022). Women and individuals who hold more flexible beliefs about gender roles tend to be more supportive of transgender athletes than men and individuals who hold traditional gender role beliefs (Flores et al., 2020; Goldbach et al., 2022). Regardless of whether the fan is male or female, sports fans are less supportive of transgender athletes than the general public (Flores et al., 2020). In general, transgender women experience more prejudice and discrimination than transgender men (Read et al., 2020; Worthen, 2013), and this pattern appears to hold true for athletes as well. For example, transgender female athletes are subjected to greater scrutiny and regulation than are transgender male athletes (Fischer & McClearen, 2020). This scrutiny could be due to the fact that even among athletes who support the inclusion of their transgender peers, there is some acknowledgement that there is a competitive advantage for transgender women (Devine, 2022).

Our study responded to the need for additional research on this topic by examining attitudes toward transgender individuals using an experimental design that manipulated a description of an athlete by varying the target athletes' gender and ability level. We were interested in whether additional characteristics of the perceiver would influence perceptions. Previous research on athletes in general demonstrates that empathetic coaches elicit higher performance (Krykun et al., 2023; Sear, 2023), which has led to a call for further investigation into the role of empathy in the world of sports. We also examined belief in a just world as a predictor of perceptions, another factor that has yet to be applied to transgender athletes. Belief in a just world is associated with ideology linked to negative attitudes toward transgender athletes, such as equal opportunity, fairness, and political conservatism (Clifton & Kerry, 2022; Sappington et al., 2019). These links suggested that belief in a just world would likely predict less support of transgender athletes. We hypothesized that

1. participants would respond most negatively to the transgender female athlete performing at a high ability level than to the other transgender athletes.
2. higher levels of empathy would predict more positive attitudes toward transgender athletes.
3. a stronger adherence to belief in a just world, for self or others, would predict more negative attitudes toward transgender athletes.
4. identifying as more politically liberal would predict more positive attitudes toward transgender athletes.

We also conducted exploratory analyses on whether participants' age, education level, and sports participation related to their perceptions of transgender athletes.

Method

Participants

Participants were young adults recruited through college classrooms, social media, and word-of-mouth. We had 108 participants complete our survey; however, three participants were excluded because they failed to respond correctly to our validity check items. Thus, our final sample was 105 young adults. One participant responded to all survey items except demographic information. The mean age of participants was 27.72 (SD = 13.39). When asked to identify their gender identity, 89 participants were cisgender women, 8 were cisgender men, and 7 categorized themselves as non-binary or "other." When asked to identify their race/ethnicity, 67 participants were White, 26 were Black, 5 were Hispanic/Latinos, and the remainder were Multiracial/Other. When asked to identify their sexual orientation, 79 participants were heterosexual, 16 were bisexual, 1 was homosexual, and 8 were "other." We asked participants about their sports participation; 46 participants had never participated in sports or participated only as a child, 44 had participated during high school, and 14 had participated beyond the high school level. When asked about their educational attainment, 8 participants reported only a high school degree, 53 had some college education, 27 had a college degree, 11 had some graduate training, and 5 attained a terminal graduate degree. We asked participants what best described their political ideology; 8 participants reported being very conservative, 16 as conservative, 12 as slightly conservative, 20 as slightly liberal, 38 as liberal, and 10 as very liberal.

In the online data collection system, participants were randomly divided into four conditions with 26

participants in the transgender man/low ability condition, 25 participants in the transgender man/high ability condition, 27 participants in the transgender woman/low ability condition, and 24 participants in the transgender woman/high ability condition. This project was approved by the university's Institutional Review Board, and participants agreed to voluntarily access the survey in order to participate.

Materials and Procedure

Participants first responded to the 16-item Belief in a Just World Scale, which consists of two sub-scales (Lucas et al., 2011). One sub-scale assessed the extent to which individuals trust that people are treated fairly in the world and they deserve the outcomes that they receive. A sample item is "I feel that people generally earn the rewards and punishments that they get in this world." The second sub-scale assessed the extent to which individuals trust that they are personally treated fairly in the world and deserve the outcomes that they receive. A sample item is "I feel that I generally earn the rewards and punishments that I get in this world." Responses were made on a 5-point scale where 1 represented "strongly disagree" and 5 represented "strongly agree." The published reliabilities for the other-focused scale range from .79 to .93, and for the self-focused scale from .88 to .94 (Lucas et al., 2011). We achieved a Cronbach's alpha of .88 for the other-focused scale and .90 for the self-focused scale.

Participants were then asked to respond to the 8-item Empathy Quotient scale (Loewen et al., 2010), which measured individuals' ability to understand and empathize with the feelings of others. A sample items is "I find it easy to put myself in somebody else's shoes." Responses were made on a 5-point scale where 1 represented "strongly disagree" and 5 represented "strongly agree." The published reliability for this scale is .76 (Loewen et al., 2010), and we achieved a Cronbach's alpha of .82.

Next, participants were randomly assigned to one of four experimental conditions. In each condition, participants read a story about a transgender athlete who was on a college basketball team. The stories varied by the gender and performance level of the athlete resulting in four conditions: Transgender man with low ability on the team; Transgender man with high ability on the team; Transgender woman with low ability on the team; Transgender woman with high ability on the team. The scenarios can be seen in Appendix A. Immediately after reading the story, participants encountered three validity check items to ascertain if they read the story. They were asked the name of the athlete at birth, the sport the athlete played and the skill level of the athlete. In order to remain in the study, participants had to answer at least two of the questions correctly, and one of the correct answers had to be the skill level of the athlete.

Participants then encountered 11 researcher-created items to examine their perceptions of the transgender athletes described in their experimental condition. The items can be seen in Table 1. These items addressed participants' support of the athlete, perception that the athlete had an unfair advantage over others, belief that the athlete was a champion for transgender rights, and agreement that the athlete encountered prejudice and discrimination. We also asked the extent to which participants felt sorry for the athlete's teammates and felt that it was harder to coach transgender athletes. Responses were made on a 5-point scale where 1 represented "strongly disagree" and 5 represented "strongly agree." We then asked participants to evaluate their personal feeling about transgender athletes and to evaluate how the "average adult" felt about transgender athletes. Responses were made on a 5-point continuum that ranged from "very negatively" to "very positively."

Participants responded to six demographic items that included age, gender, race, sexual orientation, highest level of education, and political ideology. Last, we provided the following statement to participants: "This study is investigating the issue of transgender adults participating in sports that match their current identity rather than their assigned birth identity. Do you have any additional thoughts you want to share on this topic?" We provided an open-ended field with unlimited space.

Results

We ran a 2 (gender) X 2 (ability) MANOVA and found a consistent pattern of support for the transgender man with a high ability level compared to the transgender woman with a high ability level. Participants agreed the most that they would be supportive if this athlete were playing on their college team when the athlete was a transgender man with a high ability level; they were least supportive of a transgender woman with a high ability level [$F(1, 101) = 5.88$, $p = .02$]. Participants agreed the most that the athlete had an unfair advantage over other players on the team when the athlete was a transgender woman with a high ability level; they agreed the least when the athlete was a transgender man with a high ability level [$F(1, 101) = 5.99$, $p = .02$]. Similarly, participants agreed the most that the athlete was cheating another athlete out of a spot on the team when the athlete was a transgender woman with a high ability level; they agreed the least when the athlete was a transgender man with a high ability level [$F(1, 101) = 4.28$, $p = .04$]. Participants also agreed most that the teammates have to be very careful what they say around a transgender

Table 1
Means (and Standard Deviations) for Perceptions of Transgender Athlete Questions by Transgender Status (TG) and Ability Level (A)

	TG Man/ Low A	TG Man/ High A	TG Woman/ Low A	TG Woman/ High A
If this were my college, I would be supportive of this athlete participating on the team.	3.46 (1.42)	4.04 (1.27)	3.96 (1.43)	3.21 (1.41)
This athlete has an unfair advantage over the other players on the team.	2.08 (1.32)	1.92 (1.0)	2.30 (1.35)	3.38 (1.38)
This athlete has as much right to be on the team as anyone else.	3.42 (1.58)	3.64 (1.35)	4.0 (1.33)	3.67 (1.24)
Another athlete is being cheated out of a spot on the team because of this transgender athlete.	2.58 (1.45)	2.16 (1.28)	2.07 (1.44)	2.79 (1.35)
I'm sure the transgender athlete encounters prejudice and discrimination.	4.31 (1.12)	4.32 (.90)	4.56 (1.09)	4.62 (.50)
Society should accept and support transgender athletes.	3.50 (1.48)	3.76 (1.39)	4.26 (1.23)	3.88 (1.26)
Situations like these are ruining women's athletics.	2.77 (1.51)	2.40 (1.53)	2.11 (1.31)	2.29 (1.49)
I bet the teammates have to be very careful what they say around the transgender athlete.	3.42 (1.17)	3.24 (1.23)	2.63 (1.57)	3.46 (1.18)
It is harder to coach the transgender athlete than the other athletes.	2.46 (1.30)	2.44 (1.12)	1.85 (.99)	2.21 (1.25)
I feel sorry for the transgender athlete's teammates.	1.96 (1.18)	2.08 (1.12)	1.89 (1.42)	2.50 (1.35)
This athlete is a champion for transgender rights.	2.96 (1.34)	3.68 (1.15)	3.78 (1.34)	3.46 (.98)

woman with a high ability level and a transgender man with a low ability level [$F(1, 101) = 3.82$, $p = .05$]. The observed pattern shifted when participants were asked whether the athlete was a champion for transgender rights; participants agreed most when the athlete was a transgender woman with a low ability level and agreed least when the athlete was a transgender man with a low ability level [$F(1, 101) = 4.63$, $p = .03$]. The associated means and standard deviations can be seen in Table 1.

Across conditions, participants expressed agreement that the athlete had as much right to be on the team as anyone else ($M = 3.67$, $SD = 1.37$), that society should accept and support transgender athletes ($M = 3.84$, $SD = 1.35$) and that the transgender athletes encountered prejudice and discrimination ($M = 4.45$, $SD = .94$). Participants slightly disagreed that "situations like these are ruining women's athletics" ($M = 2.40$, $SD = 1.45$). Participants also slightly disagreed that they felt sorry for the transgender athletes' teammates ($M = 2.12$, $SD = 1.28$) and that coaching the transgender athlete was harder than coaching the other teammates ($M = 2.23$, $SD = 1.17$). We found no main effects for these variables.

We focused on the items on which participants provided similar responses across conditions and then examined whether participant characteristics predicted those perceptions using Pearson's correlations. Empathy did not predict perceptions of the transgender athletes in the experimental stories; however, the higher the participants were in empathy, the more positively they rated their overall feelings about transgender athletes [$r(104) = .20$, $p = .04$]. This correlation represents a weak effect size.

Participants' belief that the world is just for them personally did not predict their perceptions of athletes. In contrast, participants with a stronger belief in a just world for others agreed less that the transgender athletes had as much right to be on the team as anyone else [$r(105) = -.22$, $p = .02$] and that the transgender athletes encountered prejudice and discrimination [$r(103) = -.32$, $p = .001$]. These participants agreed more that they felt

sorry for the transgender athletes' teammates [$r(104) = .22$, $p = .02$] that coaching the transgender athlete was harder than coaching the other teammates [$r(103) = .24$, $p = .02$], and that situations like these are ruining women's athletics [$r(103) = .21$, $p = .03$]. The higher participants belief that the world is just for others, the more negatively they rated their overall feelings about transgender athletes. These correlations represent weak effect sizes.

Political ideology predicted all of the examined variables. The more politically liberal participants were, the more they agreed that the transgender athlete had as much right to be on the team as anyone else [$r(104) = .72$, $p < .001$], that society should accept and support transgender athletes [$r(103) = .67$, $p < .001$] and that the transgender athletes encountered prejudice and discrimination [$r(103) = .40$, $p < .001$]. The more politically liberal participants were, the more they disagreed that "situations like these are ruining women's athletics" [$r(103) = -.61$, $p < .001$], they felt sorry for the transgender athletes' teammates [$r(104) = -.53$, $p < .001$] and that coaching the transgender athlete was harder than coaching the other teammates [$r(103) = -.52$, $p < .001$]. The more politically liberal they were, the more positively they rated their overall feelings about transgender athletes [$r(104) = .75$, $p < .001$]. These correlations represent moderate to strong effect sizes.

The older participants were, the more they agreed that transgender athletes encounter prejudice and discrimination [$r(99) = -.22$, $p = .03$]. The older the participant, the less positively they felt about transgender athletes [$r(100) = -.22$, $p = .03$] and the more they agreed that transgender athletes are harder to coach than their teammates [$r(99) = .26$, $p < .01$]. The more positively participants rated their own feelings about transgender athletes, the more positively they believed the "average adult" felt about transgender athletes [$r(104) = .27$, $p < .01$]. These correlations represent weak effect sizes. Education level did not predict perceptions of transgender athletes. The extent of participants' sports participation only predicted one variable. The greater the length of sports' participation , the less participants agreed that transgender athletes face prejudice and discrimination [$r(103) = -.20$, $p = .05$], reflecting a weak effect size.

Two raters categorized the qualitative comments, with an inter-rater reliability of .96. On the optional open-ended prompt, we had 23 participants respond. Responses were categorized based on their support for transgender athletes as uncertain, supportive, not-supportive, or mixed. Of the responders, seven participants provided mixed feedback, six provided positive, six provided negative, and four expressed uncertainty. Sample response options can be seen in Appendix B.

Discussion

We hypothesized that participants would respond more negatively to the transgender female athlete especially when she was performing at a higher ability level than her teammates. This prediction was supported. When the athlete was a skilled transgender woman, participants were least supportive, perceived her as having an unfair advantage, agreed that she was cheating another athlete out of a spot on the team, and that her teammates had to be careful about what they could say in her presence. Participants had the most supportive views on these same issues for the transgender man with a high level of skill. These findings reflect the commonly held perspective that a biologically male body has unfair advantages over a biologically female body (Martin & Rahilly, 2023). A recent example is the transgender female, Lia Thomas, a Division I college swimmer who transitioned after going through male puberty; her competitive success elicited a firestorm of controversy related to the extent of her biological advantages (Shermer, 2022). Even athletes who support the inclusion of transgender athletes report agreement that cisgender women are athletically disadvantaged in comparison to transgender women (Devine, 2022). This data lends support to the limited, but growing, body of evidence documenting greater prejudice toward transgender female athletes compared to transgender male athletes. Our study provides additional confirmation that these attitudes are tied to perceptions of performance ability. The transgender female athlete who performed at the highest level elicited the most negative attitudes, reflecting participants' belief that she achieved success due to biological advantages that her teammates did not have. Similarly, the transgender male athlete in our study elicited the most support when he performed at a high skill level despite not having the advantage of going through male puberty.

Our data revealed a slightly different pattern when we asked participants whether the athlete was a champion for transgender rights. Participants agreed most that the transgender woman with low ability was a champion. In other words, this athlete was biologically expected to perform well but was not performing well. Our participants may have perceived this athlete as representing a courage and perseverance in the face of adversity that highlighted the struggle for acceptance that transgender athletes face. Our participants were least impressed with the transgender man with low ability. This situation matched social expectations (biological

disadvantage leading to lower performance), and champion status is not granted to individuals who match the norm.

It is worth noting that these differences in support level were relative. Overall, our quantitative data suggested that participants held favorable perceptions of all of the athletes. Across conditions, participants agreed that the athletes had the right to compete, should be supported, and were similar to other athletes on the team. Participants also agreed that the transgender athletes dealt with prejudice and discrimination. In other words, our participants reflect the growing support for transgender athletes seen in the public at large (Avalos et al., 2022). In addition, previous researchers documented that women tend to be more supportive of transgender athletes than men (Flores et al., 2020; Goldbach et al., 2022), and our sample was primarily women.

A second goal of our study was to examine whether perceiver characteristics influenced perceptions of transgender athletes. We hypothesized that greater empathy would predict more positive attitudes toward transgender athletes and found limited support for this prediction. Participants responded similarly to the transgender athletes in the experimental stories regardless of their empathy level. The only difference we found was when we directly asked participants how they personally felt about transgender athletes in general. The more empathetic the participant, the more positively they reported feeling. In other words, empathetic participants did not respond more favorably to transgender athletes in our scenarios than did less empathetic participants; however, more empathetic participants believed that they had the most favorable attitudes toward transgender athletes. One interpretation might be that empathetic individuals are not more behaviorally or attitudinally positive toward transgender athletes, they simply assume that they are because of their self-perception of being empathetic people. Another possible explanation is that empathetic individuals have a sincere concern for transgender athletes; however, in specific situations, their feelings of fairness are more salient in driving their reactions.

In contrast to empathy, belief in a just world emerged as a significant predictor of attitudes. Failing to support our hypothesis, we found that belief in a personally just word was not predictive of attitudes. However, matching our hypothesis, a stronger adherence to belief in a just world for others predicted more negative attitudes toward transgender athletes. Participants who agreed more that the world was just (for others) did not support the athletes' right to be on the team, disagreed that the athlete encountered prejudice, felt sorry for the teammates and coach, and believed that these situations are ruining women's athletics. They also directly reported that they had more negative feelings toward transgender athletes. These correlations were weak but revealed a consistent pattern. Individuals who believe the world is just tend to support the inherent fairness of the natural order and structure of the world. These individuals may have seen the transgender athletes as inappropriately modifying that natural order. The fact that attitudes about justice were more predictive than empathy supports our earlier argument that feelings of concern for these transgender athletes might exist apart from feelings of fairness and attitude agreement.

Belief in a just world is linked to conservative political ideology (Clifton & Kerry, 2022; Sappington et al., 2019), and our data revealed that political conservatism was a similar and even stronger predictor of negative attitudes toward transgender athletes. Matching our hypothesis, the more politically liberal participants were, the more they supported transgender athletes, viewed them as similar to their teammates, and felt they experienced prejudice. Our data is consistent with limited research on this topic and meets the need for additional, confirmatory evidence that political ideology that is less tied to tradition is linked to more positive attitudes (e.g., Avalos et al., 2022; Kavoura & Kokkonen, 2021).

Because data on transgender athletes is limited, we also explored whether participants' age, education level, and sports participation related to their perceptions of transgender athletes. Education level and sports participation were not predictive, with one exception. The longer individuals had played sports, the less they believed that transgender athletes face prejudice and discrimination. This outcome could reflect a protectiveness of athletes wanting other athletes or their sports to be perceived favorably. Another explanation is that more serious athletes may place more weight on other athletes' ability level instead of aspects of the other athletes' identities. In other words, athletes may prioritize a useful teammate over other factors.

Older participants agreed that transgender athletes faced prejudice and discrimination, possibly because they also agreed that they felt more negatively about transgender athletes. One factor to consider is that older adults are more likely to hold traditional or conservative views (e.g., Peterson et al., 2020), and these views are associated with more negative attitudes toward transgender individuals. The presence of transgender athletes is a relatively new social issue, and thus, young adults, like those in our study, may be more familiar with the topic and less guarded in their attitudes.

Our qualitative data reflected much of the quantitative data; however, when participants were given the opportunity to explain their feelings, the findings revealed complexity and contradictions. Over half of the responses were either not supportive of or offered mixed support for transgender athletes competing based on their gender identity. A common response started with a globally supportive comment followed by "but…" and a qualifying comment. Even among those participants who said they generally supported transgender athletes an emergent theme was a focus on the biological advantages of men over women. These comments mirror the public conversation during policy development (e.g., Fisher & McClearen, 2020), as well as the belief, even among athletes, that transgender women enjoy a competitive advantage (Devine, 2022). It might be noteworthy that the vast majority of participants chose not to respond to the prompt, which may reflect a lack of motivation or possibly a social desirability issue where participants were reluctant to explain their opinions. The missing responses might also be a sign of conflicting opinions that were difficult to explain. Of participants who responded, 18% provided responses that cited confusion or misunderstandings. The inconsistency that we found between professed support alongside perceptions of possible unfairness is not unique to our study. Goldbach et al. (2022) surveyed college athletes across the United States and found similar misunderstandings and conflicting viewpoints about transgender athletes. Devine (2022) interviewed female Olympians and, despite their general support for transgender athletes, the Olympians reported a fear of asking questions or saying the wrong thing in the presence of their transgender peers.

The public's level of support is worthy of examination as it influences policies and funding (Martin & Rahilly, 2023; Moyer, 2022; Posbergh, 2023). Social support also affects the transgender athletes themselves. Pockets of vulnerability for transgender athletes include misuse of pronouns, discomfort in the locker room, and lack of mental health resources (Morris & Van Raalte, 2016). Transgender athletes report high levels of stress and fear of self-disclosure (Herrick et al., 2020; Walker et al., 2018). In a recent meta-analysis, Chan et al. (2024) reported that transgender athletes who face discrimination disproportionately present with mental health issues.

Our study had limitations that can be addressed by future researchers. Our participants were primarily women and politically liberal, which may have been a contributing factor to the generally positive attitudes we reported. We did not find outcomes predicted by education level, which may be an accurate finding or may reflect the fact that most of our participants had a college or higher degree. In addition to sampling broader demographics, future researchers may want to focus on athletes' perceptions of transgender athletes with varied levels of ability. Another avenue of research might be to explore whether the transgender athlete's affiliation with an individual versus team sport might impact perceptions of the athlete.

In sum, we found that participants expressed general support for transgender athletes and an awareness that these athletes face prejudice and discrimination. The transgender man with a high level of ability elicited the most support, while the transgender woman with a high level of ability elicited the lowest levels of support. As supported by our qualitative data, these findings reflect a focus on the biological advantages that accompany male puberty. Participants who believed in a just world, along with those who were politically conservative, lower in empathy, and older, were less supportive than their counterparts. These findings suggest that perceptions of transgender athletes are based on the athletes' gender identity and performance, as well as characteristics of the perceiver. An enhanced understanding of the public's perception may be useful as transgender athletes are currently at the center of controversy, and corresponding policies are in flux

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Appendix A

Experimental Conditions Varied by Gender Status (TG) and Ability Level (A)

TG Man Narrative

Mike was born a female. His given birth name was Molly.

Even before puberty started, Mike recognized that he did not feel like other teen-age girls. He questioned his identity on a daily basis. Despite these feelings, he was a star on the high school girls' basketball team and on female travel teams for basketball. In college, Mike started to see a psychologist and eventually decided to transition to being a man. He took estrogen and drugs to suppress testosterone and other male hormones. Two years after Mike transitioned, he tried out for the men's basketball team on his college campus.

TG Man/ Low A

He made the team but was not one of the better players on the team. He spent most of the season sitting on the bench. Mike was happy to be on the team, but he felt discouraged by sitting on the bench and by the rude comments he sometimes heard other players say behind his back.

TG Man/ High A

He made the team and worked his way to being a starting guard. He got more playing time than some of the players who had been on the team before he arrived. Mike was happy to be on the team, but he felt discouraged by the rude comments he sometimes heard other players say behind his back.

TG Woman Narrative

Molly was born a male. Her given birth name was Mike.

Even before puberty started, Molly recognized that she did not feel like other teen-age boys. She questioned her identity on a daily basis. Despite these feelings, she was a star on the high school boys' basketball team and on male travel teams for basketball. In college, Molly started to see a psychologist and eventually decided to transition to being a woman. She took testosterone and drugs to suppress estrogen and other female hormones. Two years after Molly transitioned, she tried out for the women's basketball team on her college campus.

TG Woman/ Low A

She made the team but was not one of the better players on the team. She spent most of the season sitting on the bench. Molly was happy to be on the team, but she felt discouraged by sitting on the bench and by the rude comments she sometimes heard other players say behind her back.

TG Woman/ High A

She made the team and worked her way to being a starting guard. She got more playing time than some of the players who had been on the team before she arrived. Molly was happy to be on the team, but she felt discouraged by the rude comments she sometimes heard other players say behind her back.

Appendix B

Sample Qualitative Comments to Open-Ended Prompt

Mixed Support (*n* = 7)

I support transgender athletes however there is a biological advantage that has to be acknowledged. Biological males have more muscle mass than females and thus have physical advantage that should be acknowledged.

I feel like I 100% support transgender athletes. My only issue is that I don't know the make up of how fair it actually is. All I see is stuff on Facebook about how it is so unfair for a transgender woman to compete in women's sports. If I knew more, I would have better answers.

Supportive (*n* = 6)

They are humans. They have a right to play sports. People saying they shouldn't do not care about human beings and suck.

I just think that people should respect other people's identities, whether they are cisgender or transgender because it doesn't effect [sic] anybody other than the person themselves. In other words, you do you.

Not Supportive (*n* = 6)

It's a difficult topic because men's bodies are built different than women's. It seems unfair for a man's body to be able to participate in a woman's sport.

I think we should be kind to transgender individuals just like any other person no matter how they identify or the choices they make. However, there isa biological difference no matter how they choose to identify and I feel strongly that biology should determine if they play on the men or women's team. I don't think they shouldn't be allowed to play, but I think based on their genetics and birth given gender, they should play on that team to keep the game "fair."

Uncertain (*n* = 4)

I feel there needs to be more education concerning the differences in transgender people's bodies when they have or have not gone through the puberty of their assigned sex. I feel conflicted.

I am confused.